Neural Networks as Parameterizers of Probability Distributions — Problems

ELG 5218 – Uncertainty Evaluation in Engineering Measurements and Machine Learning

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Part A: Conceptual Questions

A1. Universal Approximation and Distribution Parameterization

Question. Explain why a feedforward neural network can serve as a parameterizer of a probability distribution. What role does the universal approximation theorem play, and what additional step is required to ensure outputs are valid distribution parameters?

A2. Constraint Mappings for Valid Outputs

Question. A neural network outputs raw real-valued vectors. For each of the following distribution parameters, state the appropriate output transformation and explain why it is needed:

(a) Mixture weights $\{\pi_k\}$ for a K-component mixture,

(b) Variance $\sigma^2(x)$ for a Gaussian head,

(c) Full covariance matrix $\Sigma(x)$ for a multivariate Gaussian.

A3. Conditional vs. Generative Modeling

Question. Distinguish between conditional density estimation and generative modeling as two uses of neural network distribution parameterization. For each, describe what quantity the network outputs and how a sample is obtained.

A4. Mixture Density Networks (MDNs)

Question. A Mixture Density Network models $p_\theta(y \mid x)$ as a K-component Gaussian mixture. Identify all quantities the network must output, state the constraint each must satisfy, and explain why MDNs are preferable to a single Gaussian head when the true conditional distribution is multi-modal.

A5. Autoregressive Factorization

Question. State the probability chain rule factorization that underlies autoregressive models. How does an AR(2) neural network use this factorization, and what advantage does replacing the linear AR(2) mapping with a neural network offer over the classical linear model?

A6. Latent-Variable Models and VAEs

Question. In a Variational Autoencoder (VAE), two neural networks are trained: an encoder and a decoder. Describe the probabilistic role of each. Why is the true posterior $p_\theta(z \mid x)$ intractable, and what quantity does the VAE optimize instead?

A7. Learning Variance Without Variance Labels

Question. In a Gaussian mean–variance (MVE) network, the model learns $\sigma(x)$ from data even though no explicit variance labels are provided. Explain conceptually how the Gaussian negative log-likelihood loss provides implicit supervision for the variance head.

A8. Pathology: Variance Absorbing Mean Error

Question. Describe the failure mode that can arise when a Gaussian-head network is trained jointly on mean and variance from the start. What is the practical consequence, and what two remedies does the literature recommend?

Part B: Mathematical Derivations

Assume a Gaussian-head (MVE) model:

$$y \mid x \;\sim\; \mathcal{N}\big(\mu_\theta(x),\; \sigma_\theta(x)^2\big),$$

where the network outputs $\big(\mu_\theta(x), \log \sigma_\theta(x)\big)$.

B1. Gaussian Negative Log-Likelihood

Problem. Starting from the Gaussian probability density, derive the per-sample negative log-likelihood loss $\mathcal{L}(x, y)$ as a function of $\mu_\theta(x)$ and $\sigma_\theta(x)$. Show that it decomposes into two terms and interpret each term.

B2. Optimal Variance at a Fixed Input

Problem. Fix the mean $\mu(x) = \mu^*$ (treat it as a constant) and let $r = y - \mu^*$ be a fixed residual. Minimize the per-sample NLL with respect to $\sigma > 0$. Show that the optimal σ satisfies $\sigma^2 = r^2$, and interpret what this implies over a distribution of residuals at the same x.

B3. Gradient of the NLL with Respect to μ_θ and $\log \sigma_\theta$

Problem. Let $s = \log \sigma_\theta(x)$ (the raw network output). Rewrite the NLL in terms of s. Compute the partial derivatives $\partial \mathcal{L}/\partial \mu_\theta$ and $\partial \mathcal{L}/\partial s$. Show that the gradient with respect to μ_θ is scaled by $1/\sigma^2$ and discuss the implication: large predicted variance reduces the gradient signal for mean learning.

B4. AR(2) Training Objective

Problem. Given a time series $x_1, \ldots, x_T$ and a neural AR(2) model that outputs $\big(\mu_\theta(x_{t-1}, x_{t-2}), \log \sigma_\theta(x_{t-1}, x_{t-2})\big)$, write down the full training objective $\mathcal{L}(\theta)$ as a sum over $t = 3, \ldots, T$. Identify how training data triples are constructed from the raw time series.

B5. Calibration: Standardized Residuals

Problem. Define the standardized residual z_i for a test sample (x_i, y_i). State the distribution z_i should follow if the model is perfectly calibrated. Derive the expected empirical coverage for the 1σ and 2σ intervals. If you observe that 80% of test points fall within $\pm 1\sigma$, what does this indicate about the model's uncertainty estimates?

Part C: Parametric Analysis

C1. Effect of the Number of Mixture Components K

Question. In a Mixture Density Network with K Gaussian components, how does increasing K affect: (a) the expressiveness of the model, (b) the number of network output units required, and (c) the risk of overfitting? What happens in the limit $K \to 1$?

C2. Effect of Variance Regularization Strength λ

Question. In the MVE training algorithm, a regularization term $R = \lambda \| \log \sigma_\theta(x) \|^2$ is added to the NLL. Describe what happens to the learned $\sigma(x)$ as $\lambda \to 0$ and as $\lambda \to \infty$. What practical problem does each extreme produce?

C3. Warm-Up Duration

Question. In the two-phase training algorithm for MVE networks, Phase 1 trains only the mean head with a fixed variance. What is the consequence of (a) too short a warm-up ($T_{\text{warm}} \approx 0$) and (b) too long a warm-up where the mean head has fully converged before variance training begins? What does this suggest about the role of warm-up?

C4. Heteroscedasticity vs. Homoscedasticity

Question. Suppose the true data-generating process has constant noise: $y = f(x) + \varepsilon$, $\varepsilon \sim \mathcal{N}(0, \sigma^2)$ with σ^2 fixed. If you train a Gaussian-head network that outputs both $\mu_\theta(x)$ and $\sigma_\theta(x)$, what should the network ideally learn for $\sigma_\theta(x)$? What could cause it to learn a non-constant $\sigma_\theta(x)$ even when the true noise is homoscedastic?

Part D: Application and Notebook Questions

D1. Engineering Application – Probabilistic Load Forecasting

You are building a model to forecast electrical power demand y (MW) from weather features x (temperature, humidity, hour of day). The load distribution is known to be bimodal on weekends (low and high usage modes). You must provide prediction intervals alongside point forecasts.

(a) Explain why a standard regression model (single point output) is insufficient for this task.

(b) Would you choose a Gaussian-head (MVE) network or a Mixture Density Network? Justify your choice.

(c) Describe how you would evaluate whether the model's uncertainty estimates are well-calibrated, without having explicit uncertainty labels.

(d) The model is deployed and predictions are made at x^* corresponding to an unusual weather event never seen in training. How would you expect a well-calibrated model to behave, and what limitation does a Gaussian-head network have in detecting this out-of-distribution input?

D2. Notebook: AR(2) Neural Network

The notebook trains a neural AR(2) model on a synthetic time series using the Gaussian NLL loss.

(a) The notebook plots learned $\mu_\theta(x_{t-1}, x_{t-2})$ and $\sigma_\theta(x_{t-1}, x_{t-2})$ over time. If the true data-generating process has homoscedastic noise, what pattern should σ_θ show, and what pattern would indicate the model is overfitting the variance?

(b) Modify the training loop to implement the two-phase warm-up strategy: train with fixed $\sigma = 1$ for the first 50 epochs, then unfreeze the variance head. Compare the final NLL and calibration against training mean and variance jointly from the start. What do you observe?

(c) The notebook generates synthetic sequences by sampling $x_t \sim \mathcal{N}(\mu_\theta, \sigma_\theta^2)$ autoregressively. What could go wrong if the model has learned poor variance estimates ($\sigma_\theta \gg$ true σ)? Demonstrate this by artificially scaling σ_θ by a factor of 3 and plotting generated sequences.

(d) Replace the Gaussian conditional distribution in the AR(2) model with a 2-component Gaussian mixture. Describe the changes needed to the network architecture, the output layer, and the loss function.

D3. Comparing Distribution Families

Consider the following four scenarios. For each, select the most appropriate NN distribution family from {Gaussian head, MDN, Autoregressive, VAE} and justify your choice:

(a) Predicting the remaining useful life of a battery from sensor readings; the noise is expected to grow as the battery degrades.

(b) Modeling the joint distribution of all 28×28 pixel values of handwritten digit images for generative sampling.

(c) Estimating the distribution of robot arm joint angles given a target end-effector position (inverse kinematics problem with multiple valid solutions).

(d) Learning a compressed latent representation of sensor time series for anomaly detection.

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